

Jackson Stage 3 Virtual Dynamic-Cylinder Experiment

Official virtual runs R06-R15

Verified dataset analysis for paper preparation. Archive SHA-256:
b96d73adf949556ec80bd692b2642b7405c0eb7a7e9b59463bd60d84b3565ab3.

Executive result

All ten runs ended with NavigateToPose status SUCCEEDED, and all operator notes reported no contact with the cylinder. The success rate was 100%. Median navigation time was 13.317 s. Mean ground-truth path length was 1.888 +/- 0.039 m, mean final ground-truth position error was 0.212 +/- 0.030 m, and mean physical surface clearance was 0.074 +/- 0.016 m.

Frozen configuration

Parameter	Value
Controller	dwb_core::DWBLocalPlanner
Minimum angular speed	0.30 rad/s
Robot radius used by Nav2	0.16 m
Inflation radius	0.25 m
Cost scaling factor	10.0
XY goal tolerance	0.25 m
Yaw goal tolerance	0.40 rad

Aggregate metrics

Metric	Mean +/- SD	Median	Range
Navigation time	18.063 +/- 14.823 s	13.317 s	12.200 - 60.150 s
Recoveries	1.200 +/- 1.687 count	0.500 count	0.000 - 5.000 count
Dynamic trigger displacement	0.122 +/- 0.001 m	0.121 m	0.120 - 0.123 m
Ground-truth path length	1.888 +/- 0.039 m	1.877 m	1.858 - 1.994 m
Ground-truth final position error	0.212 +/- 0.030 m	0.208 m	0.180 - 0.291 m
Ground-truth final yaw error	22.394 +/- 1.846 deg	22.806 deg	18.367 - 24.064 deg
Nav2 feedback final position error	0.239 +/- 0.007 m	0.239 m	0.221 - 0.249 m
Physical surface clearance	0.074 +/- 0.016 m	0.081 m	0.043 - 0.089 m
Configured-footprint clearance	0.049 +/- 0.016 m	0.056 m	0.018 - 0.064 m
Obstacle-negotiation time	13.698 +/- 14.909 s	8.767 s	8.067 - 56.050 s
Angular reversals near obstacle	6.500 +/- 4.089 count	6.000 count	1.000 - 14.000 count
Global-plan side switches	2.900 +/- 7.810 count	0.000 count	0.000 - 25.000 count

Run-level results

Run	Status	Time (s)	Rec.	Side switches	Side	Path (m)	GT final error (m)	Clearance (m)
R06	SUCCESS	12.20	2	0	Lower	1.878	0.211	0.089
R07	SUCCESS	15.40	0	2	Lower	1.870	0.192	0.043
R08	SUCCESS	12.70	0	0	Lower	1.884	0.200	0.080
R09	SUCCESS	12.62	0	0	Lower	1.897	0.180	0.081
R10	SUCCESS	13.37	3	0	Lower	1.889	0.198	0.084
R11	SUCCESS	13.70	1	2	Lower	1.858	0.216	0.050
R12	SUCCESS	14.75	1	0	Lower	1.875	0.205	0.072
R13	SUCCESS	60.15	5	25	Upper	1.994	0.291	0.086
R14	SUCCESS	12.48	0	0	Lower	1.874	0.218	0.083
R15	SUCCESS	13.27	0	0	Lower	1.862	0.214	0.075

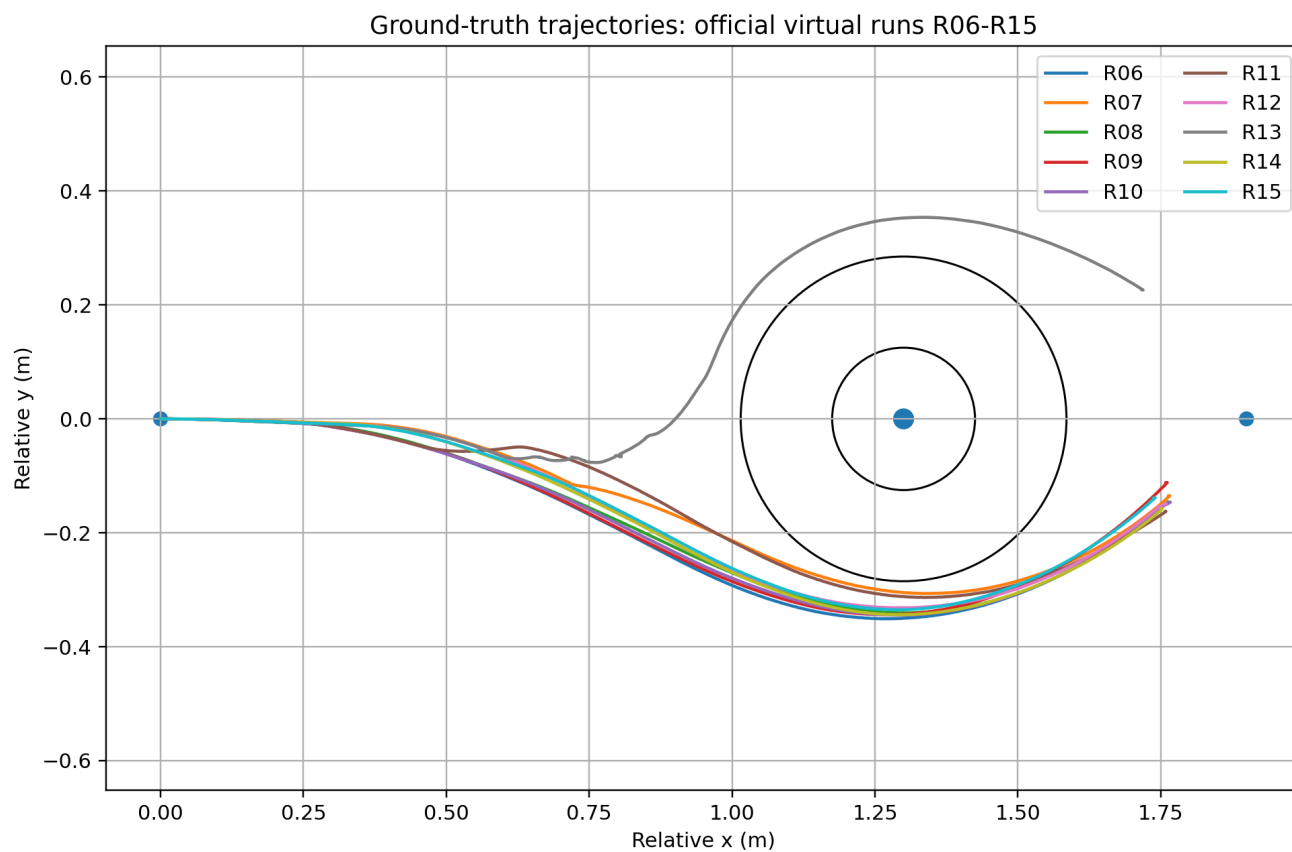
R13: valid operational outlier

R13 completed successfully and without contact, but required 60.15 s, 5 recoveries, and 25 global route-side switches. It selected the upper route, whereas the other nine runs selected the lower route. R13 is evidence of genuine planner indecision and must remain in the official sample. Both mean +/- standard deviation and median values should therefore be reported.

Trigger and safety

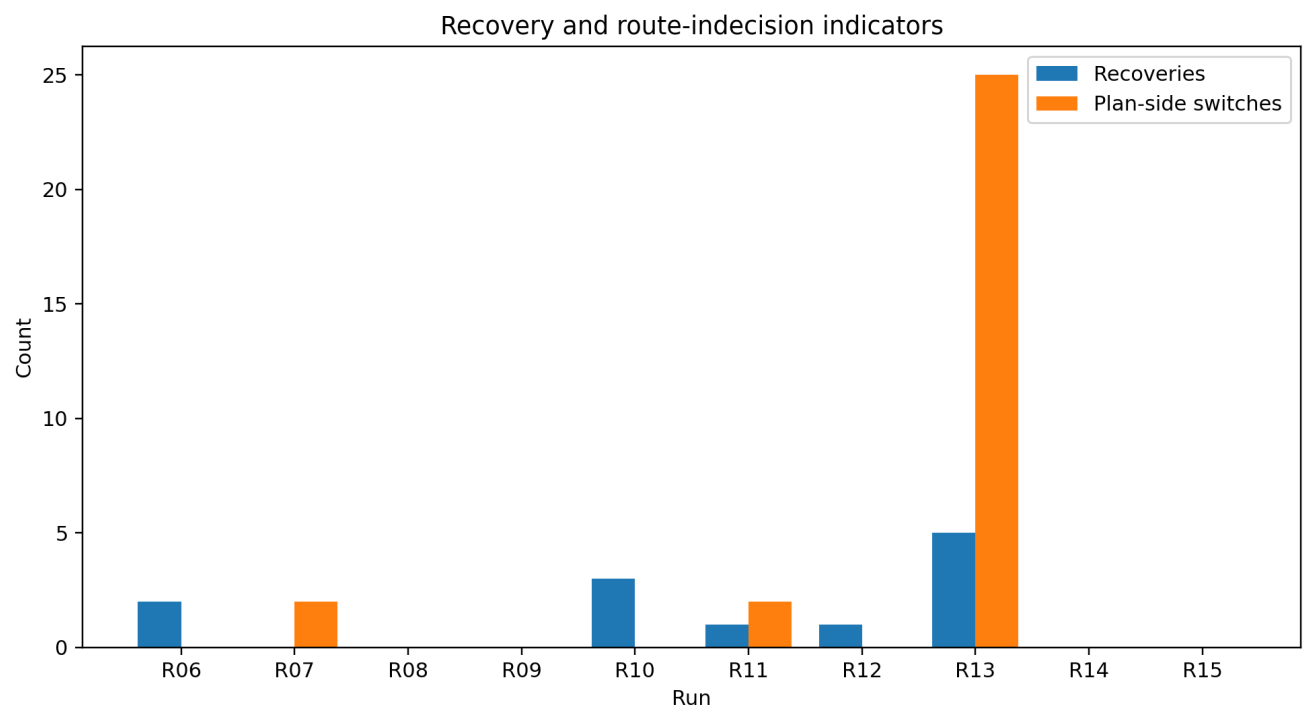
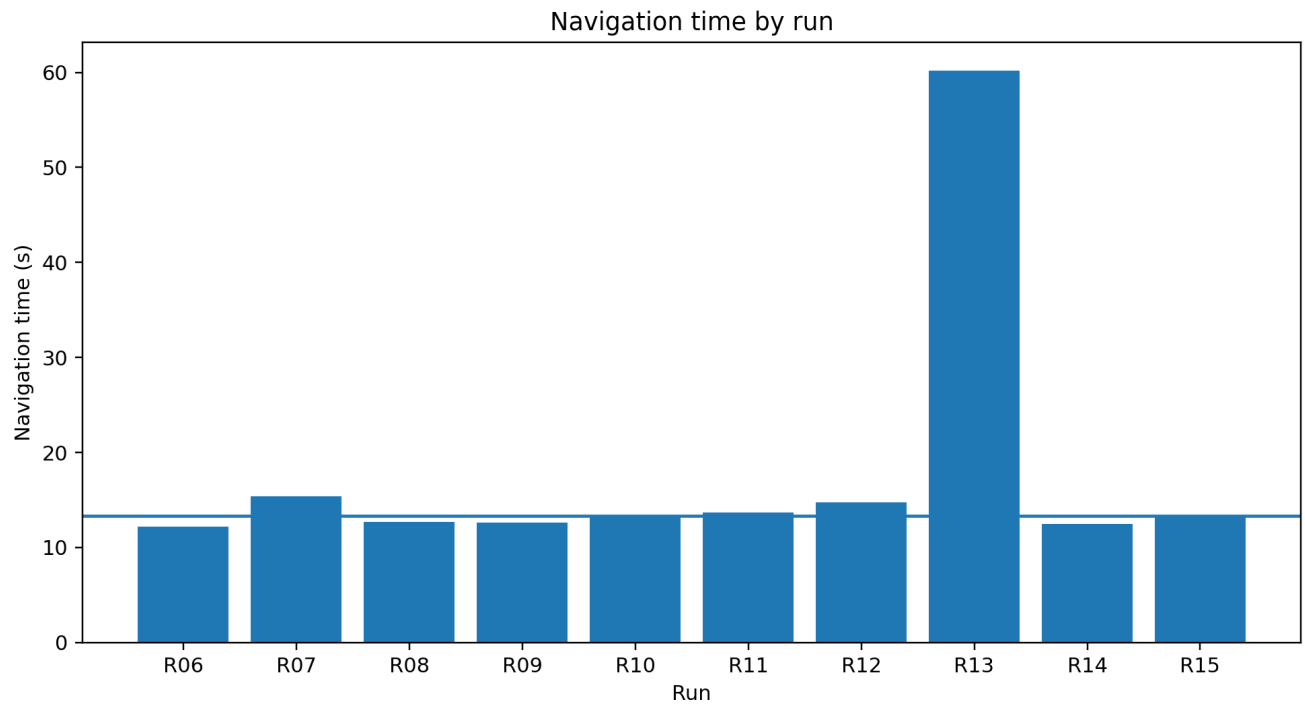
The dynamic trigger was recorded at 0.1217 +/- 0.0012 m (range 0.1201-0.1232 m). Physical surface clearance used the 0.135 m robot radius and 0.125 m cylinder radius. Every run maintained positive clearance; the minimum was 0.043 m.

Ground-truth trajectories

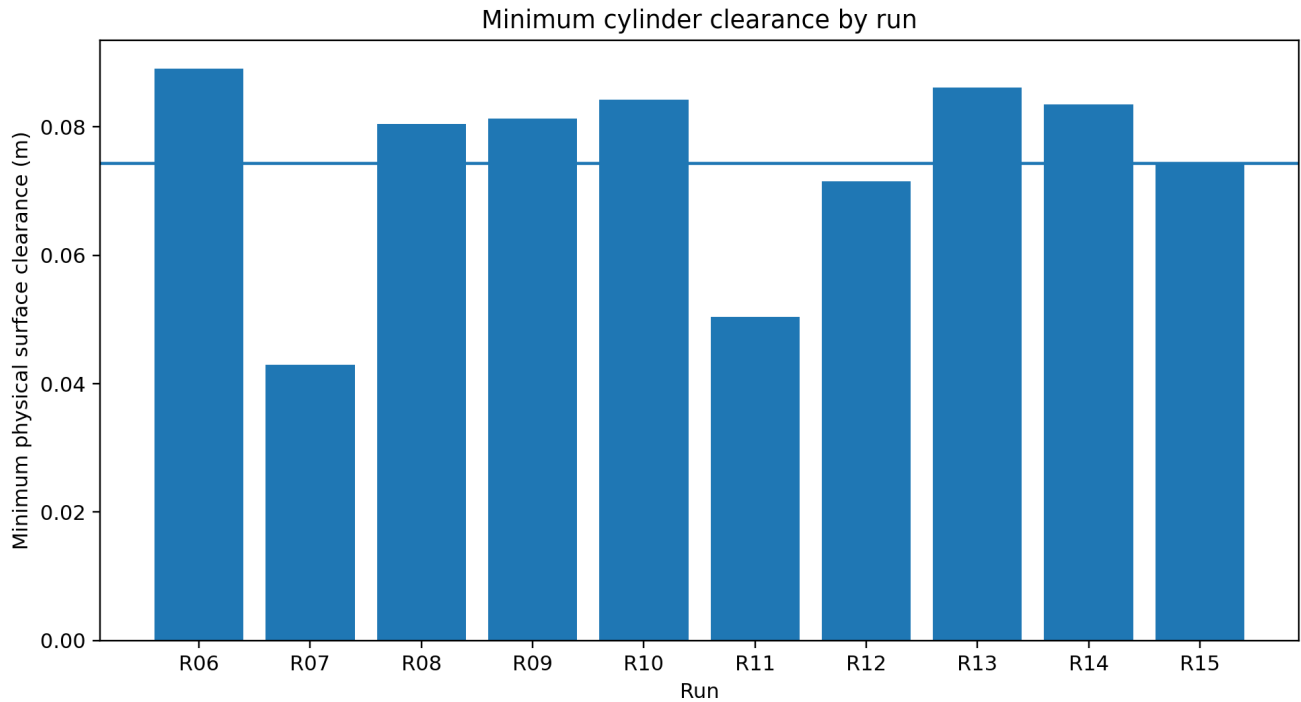


The circle at $x = 1.300$ m represents the cylinder. The larger outline represents the configured collision center-distance threshold of 0.285 m. Nine runs used the lower route; R13 used the upper route.

Timing and route indecision



Minimum clearance



Mean physical surface clearance was 0.074 m. The minimum was 0.043 m in R07. No operator-reported contact occurred.

Descriptive physical-virtual comparison

Metric	Virtual official R06-R15	Physical C1 report	Note
Duration	18.063 s mean; 13.317 s median	22.405 s mean	Navigation time versus physical motion duration
Path length	1.888 m mean	1.958 m mean	Virtual path was shorter
Final position error	GT 0.212 m; Nav2 pose 0.239 m	AMCL 0.211 m	Different pose observables
Clearance	0.074 m mean	0.122 m mean	Virtual robot passed closer

The comparison is descriptive, not inferential, because the physical and virtual reports do not use fully identical timing and error observables.

Dataset integrity

The archive checksum matched. All ten run directories were present. Every run contained one DB3 file, metadata.yaml, and a matching per-run DB3 checksum. The frozen controller and costmap parameters were identical in all runs. /map_metadata was not published, but /map and the map YAML/PGM files were preserved. This omission does not prevent analysis or invalidate the dataset.

Paper-ready result statement

In ten official virtual dynamic-obstacle trials, the robot achieved a 100% navigation success rate without contact. The cylinder was introduced at 0.1217 ± 0.0012 m of forward displacement. Mean ground-truth path length was 1.888 ± 0.039 m, mean final position error was 0.212 ± 0.030 m, and mean minimum physical clearance was 0.074 ± 0.016 m. Median navigation time was 13.317 s; one valid outlier required 60.150 s because of repeated global route-side switching and recovery behavior.